INCH-POUND

MIL-C-39006/25A

23 December 1992

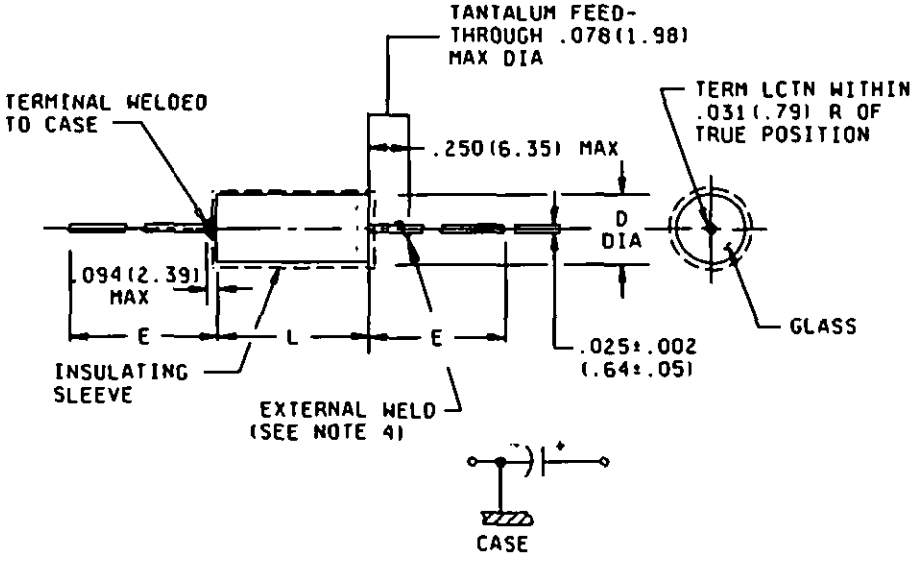
SUPERSEDING
MIL-C-39006/25
16 October 1984

MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM (POLARIZED, SINTERED SLUG) (EXTENDED RANGE), 85°C (VOLTAGE DERATED TO 125°C), ESTABLISHED RELIABILITY, STYLE CLR81

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the *Department of Defense Index of Specifications and Standards* (DODISS) specified in the solicitation: MIL-C-39006.



CIRCUIT DIAGRAM

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only.
4. The weld shall not be enclosed in the end seal.

FIGURE 1. Style CLR81 capacitors.

(A) denotes changes

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Dimensions				
Case size	Basic case		Insulated case	
	L 1/ +.031 (0.79) -.016 (0.41)	D ±.016 (0.41)	D max	E ±.250 (6.35)
T1	.453(11.51)	.188 (4.78)	.219 (5.56)	1.500(38.10)
T2	.641(16.28)	.281 (7.14)	.312 (7.92)	2.250(57.15)
T3	.766(19.46)	.375 (9.52)	.406(10.31)	2.250(57.15)
T4	1.062(26.97)	.375 (9.52)	.406(10.31)	2.250(57.15)

1/ Length of basic case sleeving shall be as specified in MIL-C-39006.

FIGURE 1. Style CLR81 capacitors - Continued.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case type: Tantalum, tubular, insulated.

Seal type: Hermetic (glass-to-tantalum anode).

(A) Terminals: Axial-wire lead: Cathode and anode (type N32, N51, or N52 of MIL-STD-1276).

Rated temperature: -55°C to +85°C, voltage derated to +125°C (see figure 2).

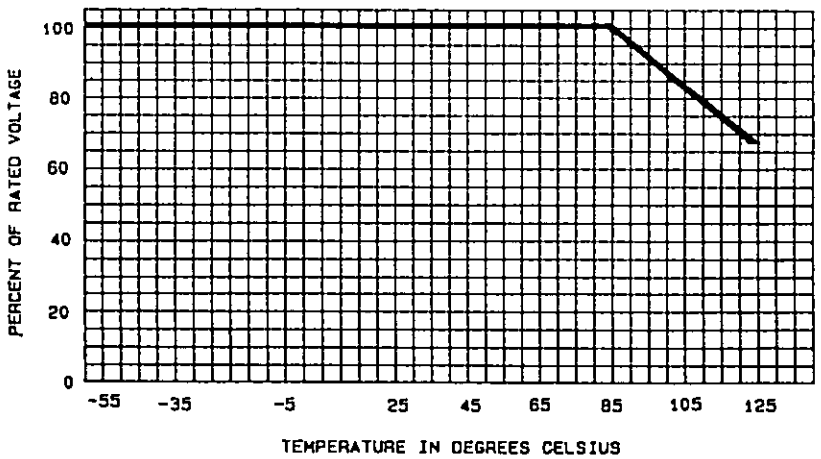


FIGURE 2. Voltage derating with temperature.

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Rated voltage: See table I.

Capacitance (Cap) (nom): See table I.

Cap tolerance: ±10 percent or ±20 percent (see table I).

FR level: M, P, R, and S (see table I).

DC leakage (DCL) (max) (at +25°C): See table I.

Cap (nominal): Method 305 of MIL-STD-202.

Dissipation factor (DF) (max): See table I.

(A) Seal: Method 112 of MIL-STD-202, conditions A or D, and C.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100 g's) or condition D (500 g's), as applicable.

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20 g's), or condition H (80 g's), as applicable.

Random vibration (when applicable): Method 214 of MIL-STD-202, test condition II-K (51 g's).

Thermal shock: Method 107 of MIL-STD-202, condition A (with step 3 at +125°C).

Number of cycles: 300 for qualification and group C; 30 for group B.

DCL: 200 percent (max) of 25°C value of table I for qualification and group C.

DCL: 125 percent (max) of +25°C value of table I for group B.

ΔCap: Within ±5 percent of initial measurement.

DF: 115 percent (max) of initial requirement.

Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hours).

Solderability: Method 208 of MIL-STD-202.

Terminal strength:

Pull: Method 211 of MIL-STD-202, condition A.

Wire-lead bend: In accordance with MIL-C-39006.

Surge voltage: See table I.

Moisture resistance: Method 106 of MIL-STD-202.

DCL: 125 percent (max) of 25°C value of table I.

ΔCap: Within ±8 percent of initial measurement.

DF: 115 percent (max) of initial requirement.

Dielectric withstanding voltage (DWV): Method 301 of MIL-STD-202. 2,000 V dc, minimum.

Insulation resistance (IR): Method 302 of MIL-STD-202, condition B. 100 megohms, minimum.

Low temperature (storage): Method 502 of MIL-STD-810.

DCL: See table I.

ΔCap: Within ±5 percent of initial measurement.

DF: See table I.

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Ⓐ TABLE I. Capacitor characteristics.

1/ Rated voltage (+85°C)	Cap.	Cap. tolerance	DC Leakage (max)		Dissipation factor (max)	2/ ESR at +25°C (max)	1/ Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max)	Capacitance change at			3/ Max 85°C 40 kHz ripple current	Case size	PIN M39006/25- 4/			
			+25°C	+85°C +125°C						-55°C	+85°C	+125°C			Failure rate level (%/1,000 hr.)			
															M(1.0)	P(0.1)	R(0.01)	S(0.001)
V dc	µF	%	µA	µA	%	ohms	V dc	V dc	ohms	%	%	%	mA rms					
6	220	±20	2	9	50	3.02	4	6.9	36	-64	+13	+16	1000	T1	0001-	0089-	0177-	0265-
6	220	±10	2	9	50	3.02	4	6.9	36	-64	+13	+16	1000	T1	0002-	0090-	0178-	0266-
6	820	±20	3	14	155	2.51	4	6.9	18	-88	+16	+20	1500	T2	0003-	0091-	0179-	0267-
6	820	±10	3	14	155	2.51	4	6.9	18	-88	+16	+20	1500	T2	0004-	0092-	0180-	0268-
6	1500	±20	5	20	172	1.55	4	6.9	18	-90	+20	+25	1600	T3	0005-	0093-	0181-	0269-
6	1500	±10	5	20	172	1.55	4	6.9	18	-90	+20	+25	1600	T3	0006-	0094-	0182-	0270-
6	2200	±20	6	24	170	1.03	4	6.9	13	-90	+25	+30	2300	T4	0007-	0095-	0183-	0271-
6	2200	±10	6	24	170	1.03	4	6.9	13	-90	+25	+30	2300	T4	0008-	0096-	0184-	0272-
8	180	±20	2	9	41	3.02	5	9.2	45	-60	+13	+16	1000	T1	0009-	0097-	0185-	0273-
8	180	±10	2	9	41	3.02	5	9.2	45	-60	+13	+16	1000	T1	0010-	0098-	0186-	0274-
8	680	±20	4	14	130	2.54	5	9.2	22	-83	+16	+20	1500	T2	0011-	0099-	0187-	0275-
8	680	±10	4	14	130	2.54	5	9.2	22	-83	+16	+20	1500	T2	0012-	0100-	0188-	0276-
8	1500	±20	5	20	170	1.50	5	9.2	18	-90	+20	+25	1600	T3	0013-	0101-	0189-	0277-
8	1500	±10	5	20	170	1.50	5	9.2	18	-90	+20	+25	1600	T3	0014-	0102-	0190-	0278-
8	2800	±20	7	28	138	1.05	5	9.2	14	-90	+25	+30	2300	T4	0015-	0103-	0191-	0279-
8	2800	±10	7	28	138	1.05	5	9.2	14	-90	+25	+30	2300	T4	0016-	0104-	0192-	0280-
10	150	±20	2	9	34	3.01	7	11.5	54	-52	+13	+16	900	T1	0017-	0105-	0193-	0281-
10	150	±10	2	9	34	3.01	7	11.5	54	-52	+13	+16	900	T1	0018-	0106-	0194-	0282-
10	560	±20	4	16	106	2.51	7	11.5	27	-77	+16	+20	1420	T2	0019-	0107-	0195-	0283-
10	560	±10	4	16	106	2.51	7	11.5	27	-77	+16	+20	1420	T2	0020-	0108-	0196-	0284-
10	1200	±20	5	20	157	1.51	7	11.5	18	-88	+20	+25	1850	T3	0021-	0109-	0197-	0285-
10	1200	±10	5	20	157	1.51	7	11.5	18	-88	+20	+25	1850	T3	0022-	0110-	0198-	0286-
10	2200	±20	7	28	142	1.01	7	11.5	15	-88	+25	+30	2300	T4	0023-	0111-	0199-	0287-
10	2200	±10	7	28	142	1.01	7	11.5	15	-88	+25	+30	2300	T4	0024-	0112-	0200-	0288-
12	100	±20	2	9	30	3.98	10	17.5	72	-44	+13	+16	900	T1	0025-	0113-	0201-	0289-
12	100	±10	2	9	30	3.98	10	17.5	72	-44	+13	+16	900	T1	0026-	0114-	0202-	0290-
12	330	±20	4	16	77	2.55	10	17.5	31	-66	+16	+20	1420	T2	0027-	0115-	0203-	0291-
12	330	±10	4	16	77	2.55	10	17.5	31	-66	+16	+20	1420	T2	0028-	0116-	0204-	0292-
12	820	±20	6	24	111	1.80	10	17.5	22	-77	+20	+25	1800	T3	0029-	0117-	0205-	0293-
12	820	±10	6	24	111	1.80	10	17.5	22	-77	+20	+25	1800	T3	0030-	0118-	0206-	0294-
12	1000	±20	8	32	95	1.55	10	17.5	17	-77	+25	+30	2300	T4	0031-	0119-	0207-	0295-
12	1000	±10	8	32	95	1.55	10	17.5	17	-77	+25	+30	2300	T4	0032-	0120-	0208-	0296-
15	68	±20	2	9	25	4.46	12	20.8	90	-40	+13	+15	850	T1	0033-	0121-	0209-	0297-
15	68	±10	2	9	25	4.46	12	20.8	90	-40	+13	+15	850	T1	0034-	0122-	0210-	0298-
15	270	±20	4	16	76	2.57	12	20.8	24	-75	+16	+19	1400	T2	0035-	0123-	0211-	0299-
15	270	±10	4	16	76	2.57	12	20.8	24	-75	+16	+19	1400	T2	0036-	0124-	0212-	0300-
15	560	±20	7	28	76	1.80	12	20.8	24	-75	+20	+25	1750	T3	0037-	0125-	0213-	0301-
15	560	±10	7	28	76	1.80	12	20.8	24	-75	+20	+25	1750	T3	0038-	0126-	0214-	0302-
15	680	±20	8	32	63	1.53	12	20.8	19	-75	+25	+30	2100	T4	0039-	0127-	0215-	0303-
15	680	±10	8	32	63	1.53	12	20.8	19	-75	+25	+30	2100	T4	0040-	0128-	0216-	0304-
20	56	±20	2	9	22	5.21	20	34.5	100	-58	+13	+15	800	T1	0041-	0129-	0217-	0305-
20	56	±10	2	9	22	5.21	20	34.5	100	-58	+13	+15	800	T1	0042-	0130-	0218-	0306-
20	220	±20	4	16	44	2.53	20	34.5	39	-60	+16	+19	1400	T2	0043-	0131-	0219-	0307-
20	220	±10	4	16	44	2.53	20	34.5	39	-60	+16	+19	1400	T2	0044-	0132-	0220-	0308-
20	470	±20	8	32	64	1.81	20	34.5	25	-62	+20	+25	1750	T3	0045-	0133-	0221-	0309-
20	470	±10	8	32	64	1.81	20	34.5	25	-62	+20	+25	1750	T3	0046-	0134-	0222-	0310-
20	560	±20	9	36	55	1.30	20	34.5	20	-62	+25	+30	2100	T4	0047-	0135-	0223-	0311-
20	560	±10	9	36	55	1.30	20	34.5	20	-62	+25	+30	2100	T4	0048-	0136-	0224-	0312-
30	33	±20	2	9	13.34	4.95	30	57.5	135	-29	+10	+12	700	T1	0049-	0137-	0225-	0313-
30	33	±10	2	9	13.34	4.95	30	57.5	135	-29	+10	+12	700	T1	0050-	0138-	0226-	0314-
30	150	±20	4	24	53.5	2.49	30	57.5	49	-42	+12	+15	1200	T2	0051-	0139-	0227-	0315-
30	150	±10	4	24	53.5	2.49	30	57.5	49	-42	+12	+15	1200	T2	0052-	0140-	0228-	0316-
30	470	±20	8	32	57	1.82	30	57.5	29	-46	+20	+25	1450	T3	0053-	0141-	0229-	0317-
30	470	±10	8	32	57	1.82	30	57.5	29	-46	+20	+25	1450	T3	0054-	0142-	0230-	0318-
30	680	±20	9	36	38	1.53	30	57.5	22	-46	+25	+30	1900	T4	0055-	0143-	0231-	0319-
30	680	±10	9	36	38	1.53	30	57.5	22	-46	+25	+30	1900	T4	0056-	0144-	0232-	0320-
60	27	±20	2	9	10.52	5.01	40	69.0	144	-24	+10	+12	700	T1	0057-	0145-	0233-	0321-
60	27	±10	2	9	10.52	5.01	40	69.0	144	-24	+10	+12	700	T1	0058-	0146-	0234-	0322-
60	100	±20	4	20	54	2.52	40	69.0	54	-36	+12	+15	1100	T2	0059-	0147-	0235-	0323-
60	100	±10	4	20	54	2.52	40	69.0	54	-36	+12	+15	1100	T2	0060-	0148-	0236-	0324-
60	220	±20	8	32	59	1.84	40	69.0	29	-40	+16	+20	1400	T3	0061-	0149-	0237-	0325-
60	220	±10	8	32	59	1.84	40	69.0	29	-40	+16	+20	1400	T3	0062-	0150-	0238-	0326-

See footnotes at end of table.

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(A) TABLE I. Capacitor characteristics - Continued.

1/ Rated voltage (+85°C)	Cap.	Cap. tolerance	DC Leakage (max)		Dissipation factor (max)	2/ ESR at +25°C (max)	1/ Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max)	Capacitance change at			3/ Max 85°C 40 kHz ripple current	Case size	PIN M39006/25- 4/			
			+25°C	+85°C +125°C						-55°C	+85°C	+125°C			Failure rate level (%/1,000 hr.)			
															M(1.0)	P(0.1)	R(0.01)	S(0.001)
V dc	µF	%	µA	µA	%	ohms	V dc	V dc	ohms	%	%	%	mA rms					
60	270	±20	9	36	27	1.33	40	69.0	23	-45	+20	+25	1850	T4	0063-	0151-	0239-	0327-
60	270	±10	9	36	27	1.33	40	69.0	23	-45	+20	+25	1850	T4	0064-	0152-	0240-	0328-
75	55	±20	3	15	8.5	5.13	50	86.5	157	-19	+10	+15	600	T1	0065-	0153-	0241-	0329-
75	55	±10	3	15	8.5	5.13	50	86.5	157	-19	+10	+15	600	T1	0066-	0154-	0242-	0330-
75	85	±20	4	22	15.5	5.46	50	86.5	63	-30	+12	+15	1000	T2	0067-	0155-	0243-	0331-
75	85	±10	4	22	15.5	5.46	50	86.5	63	-30	+12	+15	1000	T2	0068-	0156-	0244-	0332-
75	180	±20	9	36	54.4	1.80	50	86.5	30	-35	+16	+20	1300	T3	0069-	0157-	0245-	0333-
75	180	±10	9	36	54.4	1.80	50	86.5	30	-35	+16	+20	1300	T3	0070-	0158-	0246-	0334-
75	220	±20	10	40	37.0	2.23	50	86.5	54	-40	+20	+25	1800	T4	0071-	0159-	0247-	0335-
75	220	±10	10	40	37.0	2.23	50	86.5	54	-40	+20	+25	1800	T4	0072-	0160-	0248-	0336-
100	10	±20	3	12	4.5	5.97	65	115.0	200	-17	+10	+12	800	T1	0073-	0161-	0249-	0337-
100	10	±10	3	12	4.5	5.97	65	115.0	200	-17	+10	+12	800	T1	0074-	0162-	0250-	0338-
100	39	±20	5	24	10.4	3.54	65	115.0	80	-20	+12	+15	1300	T2	0075-	0163-	0251-	0339-
100	39	±10	5	24	10.4	3.54	65	115.0	80	-20	+12	+15	1300	T2	0076-	0164-	0252-	0340-
100	68	±20	10	40	11.3	2.51	65	115.0	40	-30	+14	+16	1600	T3	0077-	0165-	0253-	0341-
100	68	±10	10	40	11.3	2.51	65	115.0	40	-30	+14	+16	1600	T3	0078-	0166-	0254-	0342-
100	120	±20	12	48	25	5.76	65	115.0	30	-35	+15	+17	2000	T4	0079-	0167-	0255-	0343-
100	120	±10	12	48	25	5.76	65	115.0	30	-35	+15	+17	2000	T4	0080-	0168-	0256-	0344-
125	6.8	±20	3	15	9.0	11.71	85	144.0	300	-14	+10	+12	700	T1	0081-	0169-	0257-	0345-
125	6.8	±10	3	15	9.0	11.71	85	144.0	300	-14	+10	+12	700	T1	0082-	0170-	0258-	0346-
125	27	±20	5	24	7.5	3.54	85	144.0	90	-18	+12	+15	1200	T2	0083-	0171-	0259-	0347-
125	27	±10	5	24	7.5	3.54	85	144.0	90	-18	+12	+15	1200	T2	0084-	0172-	0260-	0348-
125	47	±20	10	40	7.9	5.63	85	144.0	50	-26	+14	+16	1500	T3	0085-	0173-	0261-	0349-
125	47	±10	10	40	7.9	5.63	85	144.0	50	-26	+14	+16	1500	T3	0086-	0174-	0262-	0350-
125	82	±20	12	48	17.4	2.82	85	144.0	32	-30	+15	+17	1900	T4	0087-	0175-	0263-	0351-
125	82	±10	12	48	17.4	2.82	85	144.0	32	-30	+15	+17	1900	T4	0088-	0176-	0264-	0352-

1/ Reverse voltage rating at 85°C is 3 V dc, and at 125°C is 2 V dc.
2/ Maximum ESR is calculated by the following equation:

$$ESR\ (max) = \frac{DF}{2\pi f C}, \text{ where:}$$

DF = Maximum dissipation factor from table I;
f = 120 Hz;
C = Nominal capacitance.

3/ For ripple current limits at various temperatures, voltages, and frequencies, see table II.
4/ Dash number shall include the letter "H" to indicate the optional vibration and shock requirements (i.e., 51 g's random vibration, 80 g's sinusoidal vibration, and 500 g's shock) or "-" will be deleted.

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TABLE II. Ripple current multipliers versus frequency, temperature, and applied peak voltage. 1/ 2/ 3/ 4/ 5/

Frequency of applied ripple current		120 Hz				800 Hz				1 kHz				10 kHz				40 kHz				100 kHz			
		Temp °C				Temp °C				Temp °C				Temp °C				Temp °C				Temp °C			
Ambient still air		≤55°	85°	105°	125°	≤55°	85°	105°	125°	≤55°	85°	105°	125°	≤55°	85°	105°	125°	≤55°	85°	105°	125°	≤55°	85°	105°	125°
% of 85°C rated peak voltage	100%	.60	.39	---	---	.71	.43	---	---	.72	.45	---	---	.88	.55	---	---	1.0	.63	---	---	1.1	.69	---	---
	90%	.60	.46	---	---	.71	.55	---	---	.72	.55	---	---	.88	.67	---	---	1.0	.77	---	---	1.1	.85	---	---
	80%	.60	.52	.35	---	.71	.62	.42	---	.72	.62	.42	---	.88	.76	.52	---	1.0	.87	.59	---	1.1	.96	.65	---
	70%	.60	.58	.44	---	.71	.69	.52	---	.72	.70	.52	---	.88	.85	.64	---	1.0	.97	.73	---	1.1	1.07	.80	---
	66 2/3%	.60	.60	.46	.27	.71	.71	.55	.32	.72	.72	.55	.32	.88	.88	.68	.40	1.0	1.0	.77	.45	1.1	1.1	.85	.50

1/ At 125°C the rated voltage of the capacitors decreases to 66 2/3 percent of the 85°C rated voltage.
2/ The peak of the applied ac ripple voltage plus the applied dc voltage must not exceed the dc voltage rating of the capacitor, either forward or reverse.
3/ The ripple current listed in table I represents a rating calculated by using a maximum internal temperature rise (ΔT) of 50°C at 40 kHz and at 85°C ambient temperature, with a maximum peak rated voltage of 66 2/3 percent of the 85°C peak voltage rating.
4/ The maximum allowable internal temperature rise (ΔT) decreases linearly to a calculated 10°C rise at 125°C ambient.
5/ The internal temperature rise is directly proportional to the ESR of the capacitor, and ESR increases with decreasing frequency.

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Stability at low and high temperatures:

Step 1 (+25°C):

DCL: See table I.

ΔCap: Within tolerance of table I value.

DF: See table I.

Step 2 (-55°C):

Impedance: See table I.

ΔCap: See table I.

Step 3 (+25°C):

DCL: See table I.

ΔCap: Within ±5 percent of step 1 value.

DF: See table I.

Step 4 (+85°C):

DCL: See table I.

ΔCap: See table I.

DF: See table I.

Step 5 (+125°C):

DCL: See table I.

ΔCap: See table I.

DF: See table I.

Step 6 (+25°C):

DCL: See table I.

ΔCap: Within 5 percent of step 1 value.

DF: See table I.

Reverse voltage: As specified in MIL-C-39006.

DCL: See table I.

ΔCap: Within tolerance of table I value.

DF: See table I.

Life at 85°C: Method 108 of MIL-STD-202.

2,000 hours (qualification):

DCL: See table I.

ΔCap: Within ±10 percent of initial measurement.

DF: See table I.

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10,000 hours (group B) and extended life:

DCL (85°C): 125 percent (max) of table I value.

DCL (25°C): See table I.

ΔCap: Within +10, -20 percent of initial measurement.

DF: 200 percent (max) of table I value.

DWV: 2,000 V dc, minimum.

IR: 100 megohms, minimum.

AC ripple at 85°C (qualification and group C):

Test ripple current: See table I.

Test voltage: 66.67 percent of rated voltage (see table I).

DCL: See table I.

ΔCap: Within ±10 percent of initial measurement.

DF: See table I.

Life at 125°C: Method 108 of MIL-STD-202.

DCL: See table I.

ΔCap: Within ±10 percent of initial measurement.

DF: See table I.

DWV: 2,000 V dc, minimum.

IR: 100 megohms, minimum.

(A) 2,000 hours (group C):

DCL (125°C): 125 percent (max) of table I value.

DCL (25°C): See table I.

ΔCap: Within +10 percent, -20 percent of initial measurement.

DF: 200 percent (max) of table I value.

DWV: 2,000 V dc, minimum.

IR: 100 megohms, minimum.

Barometric pressure: Method 105 of MIL-STD-202, condition E (150,000 feet) (45,720.1 m).

Test voltage: Rated voltage.

(A) Resistance to soldering heat: Method 210 of MIL-STD-202, condition G.

DCL: See table I.

ΔCap: Within ±5 percent of initial measurement.

DF: See table I.

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(A) Marking: In accordance with MIL-C-39006, with the following exception: The manufacturer may mark the T1 case size "H" designated capacitance as follows:

EXAMPLE: M39006/
25H0001
JAN TM
12345
9415A

Voltage groups (85°C rated voltage):

I.....6 V through 15 V

II.....25 V through 75 V

III.....100 V and 125 V

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR
Air Force -17, 99
DLA - ES

User activities:
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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